

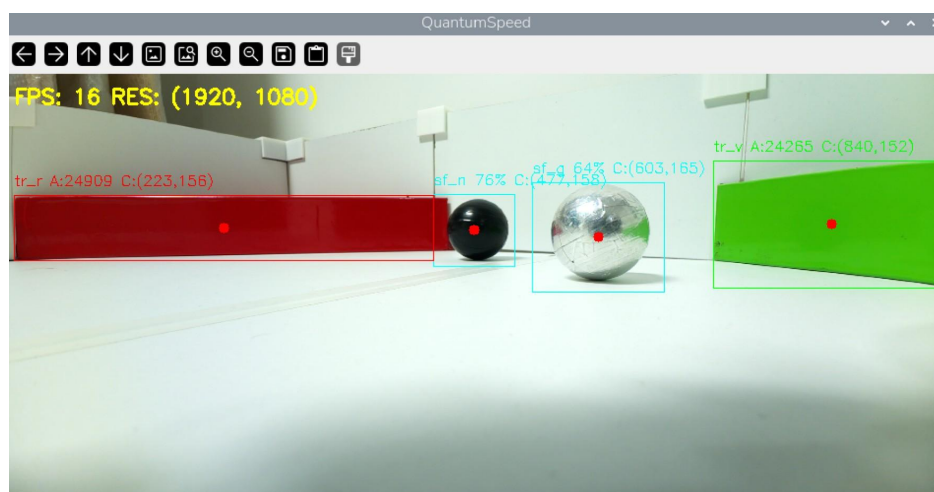
SOFTWARE

Line-following

```
def PID(cxb, centroX, cyb):  
    baseSpeed = 100  
    position = cxb  
    error = (centroX - position)*2  
    p = error  
    i = i + error  
    d = error - lastError  
    y = max(0, cyb - high * pBlack * 0.55) * 2  
    e = y*2.11 + KY + 1  
  
    lastError = error  
    motorSpeed = int((p * KP + i * KI + d * KD) * e)  
    motorSpeedx = int((baseSpeed + motorSpeed) * 0.5)  
    motorspeeddx = int((baseSpeed - motorSpeed) * 0.5)
```

To follow the line, we implement a PID algorithm, a negative feedback system widely used in automatic control systems. We detect the black line using an HSV threshold and calculate its center of mass. Next, we determine its distance from the camera's center. Based on the difference along the x-axis, the PID function calculates the required velocity variation for the wheels. This variation is then multiplied using the difference along the y-axis (see image); this approach allows us to follow the line smoothly and reliably. We also detect other colors using HSV thresholds. When red is detected, the robot stops instantly, while green tells it which direction to turn. To recognize intersections, we also utilize concentric circles: if they intersect the line more than twice, an intersection is detected. Finally, to recognize the silver line, we employ machine learning and calculate the difference in reflected light with the LEDs turned on and off.

Evacuation Zone



For object recognition inside the evacuation room, we implemented an integrated computer vision pipeline. Sphere detection is performed using a TensorFlow Lite machine learning model, trained in Google Colab on a dataset of approximately 3,000 images. Triangle recognition, on the other hand, relies on an HSV mask-based algorithm developed with the OpenCV library. Image acquisition is handled by the Pi Camera Module 3 Wide, which provides a 120° field of view and real-time autofocus capabilities. The entire system is programmed in Python, and its performance is improved by the AI HAT+ accelerator delivering up to 13 TOPS of computing power, enabling inference speeds of around 30 FPS and ensuring highly reliable and efficient object recognition. Finally, a specially designed collection chest allows the robot to gather all three spheres simultaneously, significantly reducing the overall evacuation time and improving task efficiency.

TEAM



The team was founded in 2023 under the original name 'Unicorno colorato' (which translates to 'Colored Unicorn'). The initial team members were: Simone Bettega, Matteo Defant, Samuele Contessotto, and Luca Facchinelli. Over the years, the team underwent several lineup changes (some team members and friends can be seen in the 'nazionali 2025' image). In 2024, QuantumSpeed was established with the goal of breaking into the national top 10. That year, we achieved a second-place finish, qualifying for the European Championship in Bari (IT), where we reached the podium again in second place. This year, we decided to completely overhaul the robot's underlying technology, transitioning from traditional sensors to a camera-based system. This choice led us to win the regional competition, secure second place at the national level, and qualify for the World Championship in Korea, which will mark our final chapter in the RCJ.

Simone Bettega

AI model trainer and sistemistic expert

He developed and implemented a machine learning system for silver line detection. He also implemented the lack-of-progress recovery procedure using file service. In addition, he collaborated on improving the PID control system, contributing to more accurate and stable robot navigation. He also took care of creating the venv.

Luca Facchinelli

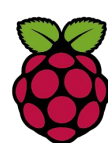
Co - Captain, Line following programmer

Luca has been part of the team since the beginning. Initially, his role was to coordinate the team, leveraging his prior experience in RoboCup Rescue Line. Over time, he specialized in C++ programming for Arduino, and over the past year, in Python for the Raspberry Pi. Currently, his primary responsibility is programming the line-following subsystem.

Matteo Defant

Captain, evacuation zone programmer and 3d modelling

Defant, the team captain, was primarily responsible for programming the evacuation zone. Since 2025, he has developed and implemented a machine learning system to recognize and classify the victims. In addition, he designs the robot in 3D using Fusion 360 and has taken care of its assembly.



RaspberryPi

At the beginning of this year, we decided to switch from sensors to a camera setup. To do this, we introduced a Raspberry Pi 5 8GB equipped with two Pi Camera 3 Wide lenses. We chose the Raspberry Pi because it is highly versatile, user-friendly, and well-suited for a wide range of applications. Furthermore, we integrated a hardware accelerator to achieve even higher performance. The Raspberry Pi handles both line-following and victim recognition. To run the entire system seamlessly, we set up a dedicated environment and configured service files, allowing the code to execute autonomously.



The entire program is built on Python, a versatile and flexible programming language known for its ease of use. Python was selected due to its widespread adoption, particularly in image processing applications. The rich ecosystem of libraries that Python offers was a decisive factor in our choice to use it for our robot. Libraries such as OpenCV and NumPy contribute significantly to our code's functionality, efficiently handling heavy computational workloads.

QUANTUM SPEED

"DA VINCI" HIGH SCHOOL (TN) - ITALY

RESCUE LINE

HARDWARE

Rescue mechanism

The victims collection system was entirely designed and 3D-printed by us. The mechanism consists of a ripper with two independent arms, so that when a victim is collected, it is then guided into the correct channel. Inside each arm, there is a microswitch to detect whether the victim has been caught. Additionally, there is a circuit that, when closed, identifies whether it is a silver, black, or fake (gray) victim. The bottom of both channels is closed by a rod connected to two other servomotors, which open only when the robot reaches the evacuation zone. In total, the mechanism consists of five servomotors and is printed using Bambu Lab's PLA Basic. The two arms are also highly practical for separating victims if they happen to be close to each other, preventing the risk of collecting two at once instead of one.

4x SERVOMOTORS

They close the gripper arms of the collection mechanism and open the exit channels. They offer an excellent power to weight ratio.

RASPBERRY PI 5 8GB

Single-board computer; after several tests, we opted for this solution due to its versatility and computing power.

2x PICAMERA MODULE 3 WIDE

Two wide-angle cameras to achieve the maximum field of view of both the line and the evacuation zone.

LOGIC LEVEL SHIFTER

Used to enable communication between the Raspberry Pi (3.3V) and the LEDs (5V).

2x LED STRIP

We use two RGBW LED strips to ensure consistent lighting, making the system independent of external illumination.

6x MICROSWITCHES

Two are used to detect whether the robot has collected the victim, while the others are used to prevent the robot from getting stuck.

2x DC-DC CONVERTERS Step Down

One step-down converter is used to step down the battery voltage from 7.4V to 5V for the Raspberry Pi, while the other powers the servomotors.

ADAFRUIT LSM6DSOX+LIS3MDL

A sensor used to measure 3-axis inclination. It is utilized for detecting slopes and ramps, as well as for executing more accurate turns.

MULTIPLEXER

An I2C multiplexer, required to expand I2C addresses and communicate with multiple sensors simultaneously.

2x OVONIC LiPo BATTERIES

We use two batteries: the first, rated at 5200mAh and 7.4V, powers the Raspberry Pi, while the second, rated at 2200mAh and 11.1V, powers the motors and sensors.

DC-DC CONVERTER Step up

One step-up converter is used to step up the battery voltage from 11.1V to 12V for the motors.

RASPBERRY PI AI HAT+

A graphics accelerator used in the victims recognition phase to increase speed and reliability.

4x GEARMOTOR POLOLU

We use four Pololu motors with a 156:1 gear ratio, powered at 12V to ensure both speed and power.

5x TOF SENSORS

They are used to bypass obstacles along the path and to navigate within the evacuation zone.

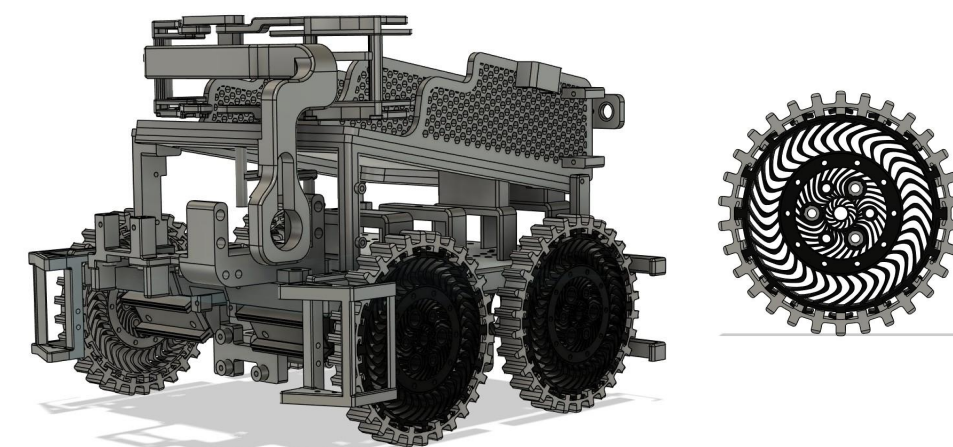
SERVO DRIVER

Required to control the servomotors directly from the Raspberry Pi.

SERVOMOTOR 25KG

It raises and lowers the collection mechanism, providing good accuracy and reliability.

Chassis and wheels



Since our first participation, we decided to custom-build the entire robot ourselves, with the exception of the sensors. The chassis was 3D printed in PLA using a Bambu Lab P1P, while the wheels have a TPU inner skeleton with a silicone coating. Additionally, all wiring and circuit boards were custom-made. This approach allows us to position all components exactly where we prefer. We placed two cameras at the front of the robot, one dedicated to line following and the other for victims recognition, ensuring that everything fits together perfectly.

